

Elementary Mathematics: Grade 5

Chapter 3: Understanding Fractions

3.1 What is a Fraction?

A fraction represents a part of a whole. It consists of a numerator and a denominator.

The number on top is called the numerator, and the number on the bottom is called the denominator.

Example: In the fraction $\frac{3}{4}$:

- 3 is the numerator (how many parts we have)
- 4 is the denominator (how many equal parts make up a whole)

Fractions can be:

1. Proper Fractions: The numerator is less than the denominator (e.g., $\frac{1}{2}$, $\frac{3}{5}$).
2. Improper Fractions: The numerator is greater than or equal to the denominator (e.g., $\frac{5}{4}$, $\frac{7}{3}$).
3. Mixed Numbers: A whole number combined with a fraction (e.g., $1\frac{1}{2}$).

3.2 Equivalent Fractions

Equivalent fractions are fractions that look different but have the same value.

You can find equivalent fractions by multiplying or dividing both the numerator and denominator by the same number.

Example:

$\frac{1}{2}$ is equivalent to $\frac{2}{4}$, $\frac{3}{6}$, and $\frac{4}{8}$.

To simplify a fraction, divide the numerator and denominator by their greatest common divisor.

Example: $\frac{4}{8} \rightarrow$ divide both by 4 $\rightarrow \frac{1}{2}$.

3.3 Adding Fractions

To add fractions with the same denominator:

1. Add the numerators.
2. Keep the denominator the same.

Example: $\frac{1}{5} + \frac{2}{5} = \frac{3}{5}$

To add fractions with different denominators:

1. Find a common denominator.
2. Convert the fractions to equivalent fractions with that denominator.
3. Add the numerators.

Example: $\frac{1}{2} + \frac{1}{4}$

- Convert $\frac{1}{2}$ to $\frac{2}{4}$
- Add $\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$